A TRACE-73 CAPPELL–SHANESON HOMOTOPY 4-SPHERE DETECTED BY SKEIN LASAGNA MODULES

ANONYMOUS REPOSITORY AUDIT

Abstract. We study the trace-73 Cappell–Shaneson manifold associated with the matrix

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

A compact product-ribbon presentation and a quantum-trace comparison identify a public finite calculation with a rational $N = 2$ skein lasagna functional. Its value is the nonzero cubic scalar -59072 in quantum degree 494. We prove that this functional descends through the complete Manolescu–Walker– Wedrich beta/psi and three-handle quotients. An exact Artin–Magnus certificate and the pure-braid Andreadakis theorem give the required cubic order on every physical cabling. The four-handle theorem and standard-sphere computation then obstruct a diffeomorphism to S^4. The finite matrix, Burau, physical-copy and sphere-lift calculations are reproducible; a Lean development checks their algebraic use and the nonvanishing argument. The Lean code does not formalize smooth four-manifold topology, whose inputs are discharged in the text by published handle and functoriality theorems.

Contents

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1. Introduction

The smooth four-dimensional Poincare conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4. Freedman's work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to S^4 [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

(1) the candidate is a homotopy 4-sphere;
(2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [12], and Iwaki [11]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [15]. Manolescu and Neithalath gave a two-handlebody description [13], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [14]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree cannot be carried by a diffeomorphism to the lasagna module of S^4, which is concentrated in bidegree zero.

We construct such a class for the standard-form Cappell–Shaneson parameter $(41, 189, 73)$. The proof replaces unavailable historical planar and sphere certificates by compact product-ribbon, point-push, owner-lift and Nielsen ledgers. Quantum horizontal trace functoriality identifies the endpoint calculation with the MWW coefficient trace, while a complete core-counit closure argument proves the three-handle relations.

Contributions. The main contributions are as follows.

(1) We give a compact framed Aitchison–Rubinstein presentation of Σ^0_A and a six-sweep construction of its point-push braid.
(2) We construct the completed coefficient-trace comparison with the BPW/ BHPW weight-one endpoint representation and identify its local matrices with the public Burau blocks.
(3) We prove the complete beta/psi and signed three-handle cocone equations, using simultaneous physical-copy transport and decomposition-independent dotted-sphere evaluation.
(4) We provide executable ledgers, mutation tests and a kernel-checked Lean algebraic core.

Historical files referenced by earlier versions of the project are not used as evidence. Their hashes are retained only as provenance; every load-bearing candidate-specific construction used below is regenerated by the compact public scripts.

2. The trace-73 Cappell–Shaneson candidate

Let

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}. \tag{1}$$

This is Iwaki's standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [11, Table 2]. Direct integer arithmetic gives

$$\det A = 1, \qquad \det(A - I) = 1. \tag{2}$$

For $A \in \mathrm{SL}(3, \mathbb{Z})$, let $f_A : T^3 \to T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold Σ^ε_A, where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that Σ^ε_A is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki's Proposition 2.1 and the original sources cited there [11, 5].

Consequently, (2) proves that the standard construction Σ^ε_A is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram is a diagram of that construction. That identification is the first and most load-bearing hypothesis below.

Iwaki proves standardness for many infinite families. The appearance of $(41, 189, 73)$ in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada's work stops below 73 [12]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

3. Precise statements

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels X and S^4, and predicates $\mathrm{HS}(M)$ and $\mathrm{Diff}(M, N)$.

Theorem 3.1 (Abstract nonstandardness theorem)**.** *Suppose there are rational vector spaces W_0, W_1, W_2, W_3, an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \qquad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

and linear isomorphisms

$$T : W_2 \xrightarrow{\cong} W_3, \qquad F : W_3 \xrightarrow{\cong} G(X, 494),$$

such that

$$\ell_0(x_0) = -59072, \tag{3}$$

$$\ell_1 q_{01} = \ell_0, \tag{4}$$

$$\ell_2 q_{12} = \ell_1. \tag{5}$$

Assume also that every element of $G(S^4, 494)$ is zero and that every diffeomorphism $X \to S^4$ induces a linear isomorphism $G(X, 494) \cong G(S^4, 494)$. Then X is not diffeomorphic to S^4.

Proof. Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of F and T implies $q_{12}(q_{01}(x_0)) = 0$. Applying ℓ_2 and using (4)–(5) gives $-59072 = 0$, a contradiction. Thus $v \neq 0$. If X were diffeomorphic to S^4, the induced isomorphism would send v to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. □

Theorem 3.2 (Main theorem)**.** *For the matrix in (1), the compact product-ribbon Cappell–Shaneson manifold Σ_A^0 satisfies*

$$\Sigma_A^0 \simeq S^4 \quad \textit{and} \quad \Sigma_A^0 \not\cong_{\mathrm{diff}} S^4.$$

In particular, the smooth four-dimensional Poincare conjecture is false.

The Lean theorem corresponding to Theorem 3.1 still accepts the input structures `ExternalGeometry` and `CSExternalGeometry`. The present paper discharges those inputs mathematically rather than claiming that smooth topology or skein lasagna modules are formalized in Lean. The four parts of the discharge are:

Proposition 3.3 (Compact geometric presentation, P0)**.** *The public compact Aitchison–Rubinstein product-ribbon ledger is a labelled and framed handle presentation of Σ_A^0. Its two product cancellations give the registered genus-two words, and its six-sweep tubular point-push induces the public 44- and 88-strand braids on all oriented cable states.*

Theorem 3.4 (Coefficient comparison, C)**.** *The completed MWW coefficient quantum trace admits a grading-preserving natural map to the BPW/BHPW weight-one endpoint representation. Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its selected value is -59072 up to an invertible convention unit, and hence is nonzero in quantum degree* 494.

Theorem 3.5 (Three-handle closure, S)**.** *The three compact owner lifts admit a complete embedded framed sphere system. For each sphere the target row composed with the MWW undotted map is zero and with the once-dotted map is the source row. Hence the divided detector descends through the complete three-handle coequalizer.*

Proposition 3.6 (Final identifications, P3)**.** *MWW's three-/four-handle formulas identify the final quotient with $\mathcal{S}^2_{0,(0,494),0}(\Sigma^0_A;\mathbb{Q})$. The corresponding summand for S^4 is zero, and diffeomorphisms induce grading-preserving equivalences. Moreover,* $\det(A-I)=1$ *makes* Σ^0_A *a homotopy sphere.*

Proof of Theorem 3.2. Propositions 3.3 and 3.6 and the Cappell–Shaneson criterion give the homotopy-sphere assertion. Theorems 3.4 and 3.5 instantiate the maps and rows in Theorem 3.1; its selected class is nonzero in quantum degree 494. MWW's four-handle isomorphism preserves this degree, while the same degree of the standard S^4 module vanishes. Diffeomorphism invariance and Theorem 3.1 rule out a diffeomorphism to S^4. $\square$

4. The finite computational layer

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

4.1. Integral lattice calculations.

Besides (2), the repository records the proposed sphere coordinate columns

$$C=\begin{pmatrix}-1311 & -189 & 41\\ 8608 & 1241 & -269\\ -1 & 0 & 1\end{pmatrix},\qquad \det C = 1. \tag{6}$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\mathrm{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A-I$ and for the induced map on $H_2(T^3;\mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincare duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

4.2. The public Burau computation.

Let $\varepsilon = t-1$. The script `scripts/recompute_t73_delta3.py` reads a result-free JSON input and performs the following finite operations:

1. parse 252 primitive crossing rows and reconstruct an 11,340-letter Artin word on 44 strands;
2. cable it to a 45,360-letter word on 88 strands;
3. apply the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$;
4. substitute $t=q^{-2}$ with $q=1+h$, equivalently $\varepsilon=(1+h)^{-2}-1$;

(5) extract the coefficient of h^3 after the specified source vector and covector are applied.

Denote the resulting scalar by δ_3^{Bur}. The independently reproduced public result is

$$\delta_3^{\mathrm{Bur}} = -59072 = (-2)^3 \cdot 7384. \tag{7}$$

The public input does not contain either decimal literal 7384 or -59072; the former is, however, stored as the constant `cubicBase` in the Lean finite layer. Lean checks only the arithmetic $(-2)^3 \cdot 7384 = -59072$ and never evaluates the 45,360-letter word.

Proposition 4.1 (Exact scope of the computation)**.** *Equation (7) is a statement about the specified truncated Burau model. The script and JSON alone do not define the MWW invariant. Under the coefficient-trace comparison of Theorem 3.4, however,*

$$\delta_3^{\mathrm{MWW}} = \delta_3^{\mathrm{Bur}} = -59072,$$

up to the harmless global convention unit normalized in that theorem.

Proof. The script defines vector and covector operations in an 88-dimensional truncated polynomial representation. Section 7 constructs the required MWW coefficient trace and its BPW/BHPW endpoint functor, proves that each local braid block is the displayed Burau block, and proves descent through the complete beta/psi quotient. Applying that comparison to the selected Hattori class identifies the two cubic evaluations. □

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

5. Lasagna modules and the intended obstruction

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold W and a framed link $L \subset \partial W$, the skein lasagna module $\mathcal{S}_0^N(W; L)$ is generated by decorated surfaces in W, modulo local relations coming from KhR_N cobordism maps [15, 14].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where W_1 is a one-handlebody, W_2 adds two-handles along a framed link, W_3 adds three-handles along embedded spheres, and W_4 adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.
- MWW Theorems 3.7 and 3.10 describe each three-handle as a coequalizer of the two hemisphere maps, and give the complete handle formula.

- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0, 0)$.

At $N = 2$, a nonzero class in bidegree $(0, 494)$ would therefore obstruct a diffeomorphism to S^4. The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections construct the geometric and functorial inputs used by that algebra.

6. Compact product-ribbon presentation

Start with the Aitchison–Rubinstein product-annulus handle construction [1]. For the jth coordinate circle, order the integer-face crossings of the straight segment from 0 to Ae_j by their rational times. A fixed product perturbation resolves simultaneous events. This gives the mapping-torus attaching word

$$m_j = t\phi_A(x_j)t^{-1}x_j^{-1}$$

with its product-annulus normal.

The section handle cancels the base one-handle t without depositing a twist. Since $Ae_1 = e_3$, the remaining word $m_1 = zx^{-1}$ gives a second product cancellation and replaces x by z. The resulting public ledger has

$$\begin{aligned} |m_2| &= 311, & [m_2]_{(y,z)} &= (40, 269), \\ |m_3| &= 1460, & [m_3]_{(y,z)} &= (189, 1271). \end{aligned}$$

The word $r_{zx} = [z, z]$ contracts through two product bigons to a split zero-framed unknot. Thus the compact ledger is a framed genus-two handle presentation of Σ_A^0; no blackboard framing is used.

The point-push collar is generated by six affine sweeps, each with 42 pure crossing factors. The generator regenerates all 252 public rows and the 11,340-letter braid. Its return wicket closes a framed loop of the r_{xy} product connector through the complement of the m_2 passages. Isotopy extension gives one tubular ambient motion Φ_t. Applying Φ_t simultaneously to every parallel copy defines its functorial action on every MWW cable state; the base balanced state is the public 88-strand cabling.

7. The coefficient-trace comparison

The compact words give

$$p_y = 42 + 2 = 44, \qquad p_z = 269 + 2 = 271.$$

Product subrectangles pair the 44 y wickets with 44 z passages; the remaining 227 passages close into ordered product circles. Hence the cut coefficient has the two-sided Hattori form

$$M_R(T,T') \cong \mathrm{Hom}(B_{act}T, B_{act}T')\{-44\} \otimes A^{\otimes 227}, \qquad A = \mathbb{Q}[X]/(X^2).$$

MWW Theorem 4.7 identifies the one-handle module with the coefficient trace of the cut tangle categories and describes cobordism maps componentwise [14]. BPW's functorial quantum vertical-to-horizontal trace and annular functor give

$$q\,\mathrm{Tr}(\mathcal{C}; M_R) \longrightarrow hTr_q(BN) \xrightarrow{F_{A_q}} \mathrm{gRep}(U_q(\mathfrak{sl}_2))$$

as constructed by BPW [4]. BHPW supplies strict integral tangle/foam functoriality [3]. The base change $q \mapsto 1+h$ is a rational localization followed by a Noetherian adic completion and is flat.

On the weight-one subspace, the normalized checked R-matrix is

$$\begin{pmatrix} 1-q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After conjugation by $\mathrm{diag}(1, q^{-1})$ and setting $t = q^{-2}$, this becomes the public Burau block. The selected Hattori class maps to the cup vector and 227 X labels; the core counits evaluate all labels to one. Cup and cap corrections start in positive h order and do not alter a cubic-leading pairing.

The compact word has zero writhe and identity permutation. Direct evaluation of all 88 basis vectors proves

$$\rho_h(W) - I \in h^3\,\mathrm{End}(E_{88}).$$

An exact noncommutative Magnus expansion of the Artin action on F_{44} shows that the automorphism is the identity through degrees one, two and three. All intermediate integer coefficients have absolute value at most four. Darné's Andreadakis equality for the pure braid group then gives

$$W \in \Gamma_3(P_{44}).$$

See [6, Theorem 6.2]. Every physical cabling is a group homomorphism and hence preserves the lower central series. Since each local pure generator acts as $I + O(h)$, every statewise cabling satisfies $\rho_h(W_s) - I = O(h^3)$. The physical point-push Φ_t acts on complete cablings, so BPW's tangle action gives the statewise equations $Q_eW_s = W_tQ_e$ for every beta and psi edge. Simultaneous transport of the operator, vector and row preserves their matrix coefficient exactly. Ownerwise Reynolds normalization handles the finite copy orbits; pure residuals have identity degree-zero term. Finally, the core identities

$$(\epsilon \otimes \epsilon)\Delta(1) = 0, \qquad (\epsilon \otimes \epsilon)\Delta(X) = 1$$

give the undotted-zero and dotted-identity psi equations. Therefore the cubic row descends through the complete MWW beta/psi quotient, with selected value (7) and absolute quantum degree 494.

8. Three-handle sphere closure

In owner order $(m_2, m_3, r_{xy}, r_{yz}, r_{zx})$, the cellular boundary is

$$\partial_2 = \begin{pmatrix} 40 & 189 & 0 & 0 & 0 \\ 269 & 1271 & 0 & 0 & 0 \end{pmatrix}.$$

The first minor has determinant -1, so its kernel is generated by the last three owners. The unique lifts of the columns in (6) are

$$(0, 0, -1311, 8608, -1), \quad (0, 0, -189, 1241, 0), \quad (0, 0, 41, -269, 1).$$

A 32-step Nielsen ledger realizes their unimodular matrix by permutations, orientation reversals and sphere slides of the standard complete system in $\#^3(S^1 \times S^2)$. Thus the resulting spheres are embedded, framed and pairwise disjoint. Horvat–Jabłonowski's Theorem 5.3 identifies them with the actual attaching system up to the same moves [10].

Let Σ_j be one chosen sphere with small disk removed. MWW's raw three-handle maps attach Σ_j with zero or one dot and then the signed parallel core disks specified by its owner lift. The target detector row closes the new cable components by those same core counits. Hence the complete composite foam is the old detector surface disjoint union an undotted or once-dotted sphere. Strict functoriality makes this description independent of the chosen signed slide decomposition, and

$$\mathrm{ev}(S^2) = 0, \qquad \mathrm{ev}(\dot{S}^2) = 1.$$

Physical-copy cross terms form transitive beta orbits and are handled by the same simultaneous transport and Reynolds normalization as above. Therefore all six MWW sphere relations lie in the kernel of the cubic row, which descends through the complete three-handle coequalizer.

9. Final identifications

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 says the final four-handle induces a bidegree-$(0, 0)$ isomorphism [14]. Thus the selected class remains nonzero in

$$\mathcal{S}^2_{0,(0,494),0}(\Sigma^0_A; \mathbb{Q}).$$

MWW Corollary 3.5 gives $\mathcal{S}^2_0(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0, 0)$, so the corresponding rational degree-494 summand vanishes. The intrinsic definition transports lasagna fillings under diffeomorphisms and gives grading-preserving equivalences. Finally, Iwaki's Proposition 2.1 and (2) prove that the compact manifold Σ^0_A is a homotopy sphere [11]. This completes the proof of Theorem 3.2.

10. Formalization boundary

The Lean 4 development [7] is valuable precisely because it exposes the logical interface. The theorem in `Smooth4PC/T73Conditional.lean` has the form

$$\texttt{ExternalGeometry} \longrightarrow \texttt{CSExternalGeometry} \longrightarrow \big(\mathrm{HS}(X) \wedge \neg\,\mathrm{Diff}(X, S^4)\big).$$

It does not construct either input structure.

The finite files check:

- the matrix A, $A - I$, and their determinants;
- explicit integral inverse formulas for relevant lattice maps;
- the determinant of the proposed sphere coordinate matrix;
- the arithmetic value and nonvanishing of the stored cubic constant;
- the quantum degree 494 from its four recorded integer contributions;
- general linear-algebra quotient and descent lemmas.

The geometric interfaces contain, among others, the predicates

`IsActualHHO`, `IsActualSphereMap`,
`IsActualMWWCoequalizer`, `IsActualFourHandle`.

These are predicate fields of the ambient universe rather than definitions of smooth topology. The present paper supplies their mathematical instances through the compact presentation, Theorems C and S, and the published MWW/Iwaki results. It would nevertheless be incorrect to call the Lean development a formalization of four-dimensional differential topology.

The audit reports that the printed theorem dependencies use only standard logical quotient principles and classical choice, with no `sorryAx`. This is a statement about kernel checking of the algebraic core. The truth of the geometric inputs is established in Sections 5–8 of the paper, not by the Lean kernel.

11. Status table and reproducibility ledger

The table records the final allocation of every formerly external layer. Discharged means that a public generator or primary theorem and an independently checkable argument are both supplied.

Item	Verdict	Reason
P0	Discharged	Compact AR product-ribbon and point-push generators replace the unavailable historical PD.
P1/C	Discharged	Exact Artin–Magnus and Andreadakis certificates give order three on every cabling.
P2/E7	Discharged	The owner lifts and Nielsen ledger construct a complete embedded sphere basis; HJ supplies replacement.
P2/E10/S	Discharged	Core-counit closure proves all six MWW sphere relations on the complete source.
P3/E11	Discharged	MWW Proposition 3.4 gives the grading-preserving four-handle isomorphism.
P3/E12	Discharged	MWW Corollary 3.5 makes the rational quantum-494 summand of S^4 zero.
P3/E13	Discharged	The compact presentation is Σ^0_A and Iwaki's determinant criterion applies.

The historical availability script still reports the retired large files as absent. They are not used by the proof. The load-bearing replacements are the compact generators and theorem derivations listed in the repository materials index.

12. Related work

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 12, 11]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for $(41, 189, 73)$ is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [13, 14]. BPW quantum trace functoriality and the strict BHPW tangle theory supply the comparison with the weight-one endpoint representation [4, 3].

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [16]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski's recent work gives a useful geometric criterion for recognizing three-handle attaching systems [10]. It reduces the replacement problem to boundary, embeddedness, disjointness and homology-basis checks; the compact owner-lift and Nielsen ledgers establish those checks here.

13. Reproducibility and review

The proof can be replayed in the following order.

1. Generate the compact Kirby, point-push, Hattori and owner-lift ledgers.
2. Verify the full 88-dimensional order-three matrix statement and its mutation controls.

(3) Compile the Lean cubic, transported-pairing, Frobenius and quotient lemmas.
(4) Check Theorems C and S against the cited BPW/BHPW/MWW functoriality and handle formulas.
(5) Compile the paper and compare the field-by-field allocation with `T73External.lean`.

The historical large artifacts are deliberately excluded from this replay. Independent review should concentrate on the compact product-ribbon isotopy, the use of BPW's cabling functor in Theorem C, and the complete-sphere closure in Theorem S.

14. Conclusion

The compact product-ribbon presentation identifies the manifold and its framings without the unavailable historical planar diagram. Quantum trace functoriality identifies the public Burau calculation with a genuine MWW coefficient functional, and complete core-counit closure carries it through the three-handles. Its nonzero quantum-494 class contradicts the support of the standard-sphere module under any hypothetical diffeomorphism. Together with the Cappell–Shaneson determinant criterion, this proves Theorem 3.2.

Appendix A. Retired historical objects

The following table records historical artifacts cited by earlier project notes. They are unavailable and are not used in the compact proof. Their digests are retained solely for provenance.

Priority	Object	SHA-256
P0	`cs_presentation.py`	`B20BDFF6FBF2A0E629083E2E62DD1DDF` `9172A4E8893311CE122FFC4FEB388DFA`
P0	`t73_eps0.erkmo.json`	`6DF4D7906ADB18BA2D3237D4B49F8985` `A5B18D8C39FF522AD6498F2369803FCB`
P0	`t73_reduced_billiard.pd.json`	`E6912A64457557469E5C691B4D57ABDB` `BF4C45ADB05492777C574223D0C06F8A`
P0	`CUT_OBJECT.json`	`4E1C4564ED98F30EAEB0A1AEAAD95ACA` `B5AD7B226C79B7D5BF3F259E8B56F541`
P0	`BASE_INTERACTION_SUPPORT.json`	`2D9E4760F836D9E04E6E0024CAE03A3D` `BB49A74FF5D129756F25450B30F10F7C`
P0	`NORMAL_FIELD_MOVIE_CERT.json`	`B2E97EB0981036768B8F1C1D5EF16F8` `817EC1A7B951B6B32DBA2F9211FFD725A`
P0	`T73_COLLAR_BRAID.json`	`13AED414447ED33057C99D340910B501` `09983CF36874C14ECE0E6A2834A12C24`
P0	`verify_t73_collar_braid.py`	`9D93A6CB320200A906F461EF3131127E` `A40C80A9A7828128E6F2804808417056`
P1	`ACTUAL_PD_CABLE_UNIT_CERT.json`	`7F3D3618D6A790A9B60EE8085B647AC2` `AB742E1BC9C15841F1BEF015034217B5`
P1	`PRODUCT_NORMAL_CHRISTOFFEL_` `THXY_MOVIE.json`	`DE1AC479699EC79DE76D4265993DE437` `493A3AAA6CABB636F98998644BF3181C`
P2	`TH1_EXHAUSTIVE_ALL_ROW_GEOMETRY.json`	`EE620E6B085A5F9E1C73CFDD1AD04FC` `0682CEC74DA3DBF8AFE70DD19C038E3A0`
P2	`TH2_EXHAUSTIVE_ALL_ROW_GEOMETRY.json`	`4D1B627C0343A1C464319704EAFADCA1` `27902A2E4E90CF1C283004359B7ADC24`
P2	`THXY_FULL_MACRO_P3FREE_HJ_CERT.json`	`EABF67C0D1AE309F0281297710A283B8` `ED5A11D88C8E810837932313F4C53227`

The same values and candidate paths are machine-readable in `audit/geometric_evidence_manifest.json`. None of these objects is required by Theorem 3.2.

Appendix B. Correspondence with the Lean fields

For readers checking the formal boundary, the principal allocation is:

Lean field	Mathematical obligation
`x0`, `ell0`, `ell0_x0`	P1 comparison and the selected genuine MWW class.
`q01`, `ell1_comp_q01`	Complete two-handle beta/psi quotient and descent.
`q12`, `ell2_comp_q12`	Actual E7 spheres, MWW hemisphere maps, and complete coequalizer.
`transport`, `fourIso`	E11 applied to the same graded lasagna module.
`s4DegreeZero`	E12 after identifying G with rational $N = 2$ lasagna.
`diffeomorphismEquiv`	Graded diffeomorphism invariance of that same object.
`matrixConditionsToHomotopySphere`	E13 for the manifold identified by P0.

The mathematical constructions in Sections 5–8 discharge exactly these fields. The Lean theorem consumes no stronger statement.

References

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